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HIP

Hip biomechanics differ in responders and non-responders to an ACL injury prevention program

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Abstract

Purpose To investigate the differences in demographic, anthropometric, biomechanical, and/or performance variables between those that do (responders) and do not (non-responders) exhibit reductions in knee abduction moments after an anterior cruciate ligament injury prevention program (ACL-IPP).

Methods Forty-three adolescent female athletes completed biomechanical (3D motion analysis of a drop vertical jump) and performance testing before and after randomization into a 6-week ACL-IPP. Participants were classified into responders and non-responders based on their level of reduction of knee abduction moment from pre- to post-test.

Results Compared to non-responders, responders exhibited increased hip adduction excursion at baseline ($p = 0.02$) and trended towards attending more training sessions ($p = 0.07$) and participating in soccer and not basketball ($p = 0.07$). Responders also showed greater improvements in hip flexion angles ($p = 0.02$) and moments ($p < 0.001$), and knee abduction angles ($p < 0.001$) and excursions ($p = 0.001$). There were no significant differences in age or experience with prior injury prevention programs (n.s.).

Conclusions After an ACL-IPP, athletes that exhibit the greatest reduction in knee abduction moments exhibit greater hip adduction excursion at baseline and show corresponding improvements in hip flexion and knee abduction kinematics and hip flexion moments. These results can help clinicians prospectively identify individuals that may not respond to an ACL-IPP and target individualized training for those at risk of injury.

Level of evidence I.

Clinical trial registration number Clinicaltrials.gov NCT02530333.

Keywords Anterior cruciate ligament · Injury prevention · Knee valgus · Knee abduction moments

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Introduction

Anterior cruciate ligament injury prevention programs (ACL-IPP) are generally successful at reducing the incidence of non-contact ACL injuries in female athletes [34, 38, 39] and provide cost-effective methods of reducing long-term medical expenditures associated with secondary knee osteoarthritis and associated disability [29]. ACL-IPPs that successfully reduce injuries typically include a series of neuromuscular training progressions, including plyometric, agility, strengthening, and balance interventions to modify biomechanical movement strategies that may place an athlete at risk for suffering an ACL injury [34]. However, controversy remains about the effectiveness of these programs,[6, 25] as only a limited number of studies have individually reported substantial reductions in injury rates [34]. Despite increasing awareness of the injury and more widespread

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implementation of IPPs, ACL injury rates remain high in multi-directional sports, with incidence rates increasing over the past 10 years in certain high-risk sports [1]. Further research to improve the effectiveness and efficiency of ACL-IPPs is warranted.

ACL-IPPs are currently inefficient when used as "one size fits all" programs, as up to 108 female athletes need to participate in an ACL-IPP to prevent just one ACL injury [28]. One currently used method to improve the efficiency of ACL-IPPs is to use effective screening measures to identify athletes at the highest risk of injury and target preventive training towards this high-risk group;[2], however, a more novel and direct approach may be to split athletes into groups of athletes that will respond positively to standard ACL-IPP (responder) and those that show no or minimal changes (non-responder) in injury risk, with more individualized preventive training administered to non-responders. While reduced injury incidence is the ultimate definition of program success, prospective randomized controlled trials require significant temporal, financial, and human resources. Since ACL–IPPs are thought to reduce injury risk by modifying high-risk biomechanical strategies, defining success by the levels of change of biomechanical risk factors may be the best paradigm.

While there is likely not one isolated movement strategy that increases ACL injury risk, prospective data indicate that the force pushing the knee into dynamic valgus (knee abduction moment) measured during a drop vertical jump (DVJ) screening task may be the best individual biomechanical predictor of injury risk [8]. The purpose of this study was to identify demographic, anthropometric, biomechanical, and/or performance variables that characterize athletes that respond favorably to a warm-up based ACL-IPP with reduced knee abduction moments (responders) compared to those that demonstrate minimal change or increase in knee abduction moments (non-responders). It was hypothesized that responders would be younger, have less experience with prior preventive programs, and demonstrate higher-risk biomechanical profiles (e.g. higher levels of dynamic lower extremity valgus) at baseline than non-responders. Findings from this study hope to improve the identification of athletes that might not show a positive response to a conventional ACL-IPP so that targeted, individualized training can be prescribed.

Materials and methods

This study was a sub-analysis of a larger study that examined sport-specific biomechanical adaptations after an ACL injury prevention program [31]. Participants were recruited from competitive basketball and soccer clubs to participate in the study, and were included if they: (1) were 13–19 years old, (2) participated in middle or high school level competitive women's sport(s) and considered either basketball or soccer as their primary sport, (3) were cleared for full participation, and (4) had no lower extremity injury at time of testing. Participants were excluded if they reported a previous lower extremity surgery within the past 6 months or had ever been previously diagnosed with a vestibular, balance, or cardiac disorder.

Enrolled participants were cluster randomized by the team into parallel intervention (ACL-IPP training) and control groups. If participants could not commit to completing the entire study, including the 6-week intervention with pre- and post-tests, they were excluded from further testing and analysis after the initial pre-test data collection. After pre-testing, participants were cluster randomized by the team into intervention (ACL-IPP) and control (continued normal sport practice and training) groups. Eighty-nine participants (intervention: $n = 43$, age = 15.4 ± 1.0 years, height = 164.7 ± 6.9 cm, mass = 59.8 ± 10.9 kg; control: $n = 46$, age = 15.6 ± 1.5 years, height = 167.1 ± 6.1 cm, mass = 60.5 ± 6.8 kg) were included in these analyses.

Training protocol

The warm-up based ACL-IPP published by Labella et al. was used as the neuromuscular training stimulus for this study (Table 1) [12]. Training sessions were performed prior to each team's practice at their own training facility and were all led by an experienced physical therapist with speciality certifications in sports and orthopaedics. The IPP leader provided real-time feedback with verbal cues to all participants, focusing mostly on soft landings and frontal plane knee control, as described in the original intervention [12]. Training sessions were held 2–3 times per week for 6 weeks, in accordance with each team's practice schedule, and lasted 20–25 min in duration. Typical warm-up based prevention programs are designed for an entire season, yet 6 weeks was chosen in this study to best align the basketball and soccer off-seasons as part of the larger study and past evidence supports the ability to observe neuromuscular adaptations within 6 weeks of the onset of training [4]. Attendance was recorded at each training session, but participants were not excluded from the study for non-compliance to best replicate current clinical practice.

Data collection

All participants completed identical testing sessions before and after the 6-week training period in a standard biomechanics laboratory setting. Participants in the intervention group were tested within 2 weeks prior to the scheduled onset of the training program and re-tested between 2 and 10 days after the completion of the last training session. Prior

Table 1 ACL IPP protocol used in this study

Exercise	Week 1	Week 2	Week 3–6
Strengthening			
Single-leg heel raises	10 reps	20 reps	30 reps
Squats	10 reps	20 reps	30 reps
Plank/side plank	10 s	20 s	30 s
Push-ups	10 reps	20 reps	30 reps
Stationary lunge	10 reps	–	–
Lateral lunge	–	20 reps	30 reps
Diagonal lunge	–	20 reps	30 reps
Walking lunge	–	–	30 reps
Supermans	10 reps	20 reps	30 reps
Swimmers	10 reps	20 reps	30 reps
Modified supermans	–	20 reps	30 reps
Plyometric			
Ankle bounces	10 s	20 s	30 s
Tuck jumps	10 s	20 s	30 s
180 degree rotate	10 s	–	–
Squat jump	10 s	20 s	30 s
Double-leg broad jump for distance	10 s	–	–
Forward/lateral cone jumps	10 s	20 s	30 s
Single-leg bound in place	10 s	–	–
Scissor jumps	–	20 s	–
Lateral bounding	–	20 s	–
Single-leg Hop, Hop, Stick	–	20 s	30 s
Broad jump × 3, vertical jump	–	20 s	–
Single-leg jump for distance	–	–	30 s
Jump into bounding	–	–	30 s
Diagonal bounding	–	–	30 s
Agility			
Shuttle run	50 ft, 10 reps	50 ft, 10 reps	50 ft, 10 reps
Diagonal run	50 ft, 10 reps	50 ft, 10 reps	50 ft, 10 reps
Lateral shuffle	15 ft, 10 reps	15 ft, 10 reps	15 ft, 10 reps

All participants completed a dynamic warm-up prior to these exercises that included jogging, carioca, side shuffle with arm swing, sprint at 75% maximum, high-knee skipping, high-knee carioca, sprint at 100% maximum, backward jog, bear crawl, butt kickers, backward jog half-length, turn and sprint, and diagonal skipping. Also 10 repetitions each of arm swings, trunk rotations, and leg swings

to testing, participant's demographics and anthropometrics were recorded and an electronic questionnaire detailing their history of previous injuries and experience in preventive training programs was completed (REDCapTM software, Version 4.14).

Biomechanical measures

Instrumentation Participants wore standardized footwear (adidas® adipure 360.2, Portland, Oregon, USA) throughout all testing sessions. For three-dimensional biomechanical analysis, participants were instrumented with 43 retroreflective markers on their trunk, and bilaterally on their upper and lower extremities as has been previously published [30]. Once instrumented, participants completed a static trial in neutral alignment, by which all subsequent biomechanical measures were referenced. Both three-dimensional kinematic data (200 Hz) via a 14-camera system (Eagle cameras, Motion Analysis Corporation, Santa Rosa, CA, USA) and kinetic data (1200 Hz) via dual, in-ground, 90 × 60 cm multi-axis force plates (AMTI, Watertown, MA, USA) were collected using Cortex software (version 5, Motion Analysis Corporation, Santa Rosa, CA, USA).

Procedures The drop vertical jump was used in this analysis because this task elicits consistent biomechanical performance [30] and may be the best task to predict subsequent knee injury [8]. To perform this task, a 31-cm box was placed directly adjacent to the force plates. Participants were instructed to stand with their feet 35-cm apart and their toes

hanging off of the front edge of the box. When ready, participants slid forward off the box with both feet at the same time, landed simultaneously with both feet and immediately performed a maximal countermovement jump while reaching for an overhead target with both hands. The overhead target was placed over the force plates at each participant's maximal reach obtained during 3–5 trials of a double-leg countermovement jump prior to data collection.

Data reduction All biomechanical data were processed with Visual 3D (Version 5, C-Motion, Inc., Rockville, MD, USA) using custom MATLAB code (Version 8.0, The Mathworks, Natick, MA). Hip joint centers were calculated using the Bell Method,[3] while the knee and ankle joint centers were defined as the centroid position of medial and lateral knee joint line and malleoli makers, respectively. Joint angle and moment data were reduced using low-pass, fourth-order Butterworth filters with a cutoff frequency of 12 Hz.

Biomechanical measures for hip flexion and adduction, and knee flexion and abduction have all been theorized to influence dynamic lower extremity valgus movement patterns and are considered high-risk of ACL injury,[8, 10] and thus were the focus of these analyses. For each of these motions, peak joint angles, total joint excursions (absolute value of the peak angle minus the angle at initial contact), and external joint moments were extracted for the first jump landing of the DVJ during the landing phase (period from initial contact to maximal descent of the body center of gravity) of each trial.

Limb dominance

To best standardize limb comparisons across the variety of participants of different ages and sports, analyses were performed on each athlete's dominant limb as defined by the side which produced the longest maximum jump distance during a forward triple hop test. Participants performed three hops on each limb, with the order of testing counterbalanced for each participant. Performance was measured as the distance between the starting line and point of toe contact upon completing the third hop. Trials were repeated if the participant lost balance, contacted the ground with their opposite limb, or hesitated between hops. Only measures from the dominant limb were analyzed in this study.

Performance measures

Three standard physical performance tests were completed at each testing session (triple hop, double-leg countermovement jump, agility *T* test). The countermovement jump was completed while reaching up towards an overhead target. Jump height was measured with motion capture as the difference in the vertical position of the center of mass between the static standing trial and the peak height during each trial.

Performance during the agility T-test was measured using a FITLIGHT Trainer™ (FITLIGHT Sports Corp., Ontario, Canada) timing system. Participants sprinted forward for 10 m, side-shuffled 5 m in one direction, changed direction to side shuffle 10 m in the opposite direction, and again changed direction for 5 m back to the middle and sprinted backwards through the finish line. Time was measured as the total time between crossing the start and finish line. Four total trials were performed, counterbalancing two trials side-shuffling to the right first, and two to the left first.

All participants provided written informed parental consent and participant assent as approved by the Institutional Review Boards at High Point University and the University of North Carolina at Greensboro.

Statistical analysis

All statistical analyses were performed in SPSS, version 24 (IBM Corp, Armonk, New York), with statistical significance set a priori at $\alpha = 0.05$. Values from the participants' dominant limb were analyzed. Delta scores were computed for all biomechanical variables (post-test–pre-test) to determine changes in each measure over time. To detect the effectiveness of reducing knee abduction moments in the entire cohort, an ANCOVA was performed to identify differences in knee abduction moment delta scores between intervention and control groups while controlling for baseline knee abduction moment values.

To identify a true positive response to the ACL–IPP, a minimal detectable change (MDC) score for knee abduction moment was calculated. The MDC identifies a threshold at which changes between scores can be confidently interpreted as true change and not random measurement error. Using a 95% confidence interval (CI) and the standard error of measurement from a previously published DVJ reliability study [30], the MDC was calculated as 9.4 N m. Thus, participants that exhibited a reduction of knee abduction moment greater than 9.4 N m from pre- to post-test were considered "responders" and all other intervention group participants were classified as "non-responders".

Statistical analyses were then performed to identify differences in demographic/anthropometric, biomechanical, performance, and sport and injury history data between responder and non-responder groups. Independent *t* tests identified differences in demographic/anthropometric variables (age, height, and weight), training sessions attended, and physical performance pre-test results and delta scores. Chi-square tests of independence were performed to compare binary categorical data, including

dominant sport, past injury history, and history of previous neuromuscular preventive training.

Biomechanical data were analyzed with four separate MANOVA models for both pre-test (to identify differences in baseline biomechanics) and delta scores (to identify concurrent biomechanical changes) including the peak angle, excursion, and peak external joint moment for (1) hip flexion, (2) hip adduction, (3) knee flexion, and (4) knee abduction. The multivariate statistical significance of each model was analyzed using Wilk's Lambda. Follow-up post-hoc pairwise comparisons of significant models were performed using independent *t* tests. Cohen's *d* effect sizes and 95% CI were calculated for all biomechanical variables that exhibited statistically significant effects. Ensemble curves of select variables were generated in MATLAB by normalizing each variable to the duration of ground contact and were used to visually compare and interpret differences between groups.

An a priori power analysis was performed to identify the appropriate sample size for the randomized controlled trial, deeming 80 participants adequate to achieve 80% power at a statistical significance criterion of 0.05 with a moderate effect size (0.38).

Results

There was no significant difference in knee abduction moment delta scores between intervention and control groups (n.s.). Descriptive statistics of demographic/anthropometric, performance and sport, injury, and training history for the responder and non-responder groups can be found in Table 2. There were no significant differences between responders and non-responders for age, height, or weight (n.s.).

Biomechanical baseline and delta scores are reported in Table 3. Differences in hip adduction biomechanics were identified between groups at baseline ($\lambda = 0.82$, $p = 0.049$) such that responders had greater levels of hip adduction excursion than non-responders ($p = 0.02$, $d = 0.82$). No other significant differences in baseline hip flexion, knee flexion, or knee abduction biomechanical measures were identified (n.s.). Significant differences were identified between non-responders' delta scores for knee abduction ($\lambda = 0.42$, $p < 0.001$) and hip flexion ($\lambda = 0.63$, $p < 0.001$). In addition to greater reductions in knee abduction moment ($p < 0.001$, $d = 3.12$), responders showed greater reduction in knee abduction peak angles ($p < 0.001$, $d = 1.66$) and excursions ($p = 0.001$, $d = 1.43$). Responders also exhibited greater increases in peak hip flexion angles ($p = 0.02$, $d = 0.80$) and peak hip flexion moments ($p < 0.001$, $d = 1.32$) than non-responders.

Table 2 Mean ± standard deviations of anthropometric, adherence, and performance data and frequencies of dominant sport, history of injury and history of prior preventive training

	Responders ($n = 8$)	Non-responders ($n = 34$)	*p* value
Anthropometrics			
Age (years)	15.5 ± 1.1	15.5 ± 0.9	n.s
Height (cm)	165.3 ± 4.1	164.8 ± 7.4	n.s
Mass (kg)	62.7 ± 10.3	59.0 ± 11.2	n.s
Training attendance			
Training sessions	12.3 ± 1.8	10.0 ± 3.3	n.s
Performance tests			
CMJ Height, pre (cm)	41.3 ± 5.1	41.3 ± 5.7	n.s
CMJ Height, delta (cm)	− 0.003 ± 0.02	− 0.004 ± 0.017	n.s
Triple Hop, pre (cm)	162.2 ± 21.6	165.3 ± 20.5	n.s
Triple Hop, delta (cm)	1.50 ± 24.1	− 1.14 ± 15.7	n.s
Agility *T* test, pre (s)	15.3 ± 2.1	14.9 ± 1.9	n.s
Agility *T* test, delta (s)	− 0.54 ± 1.0	− 0.46 ± 1.8	n.s
Sport and injury history			χ^2, *p* value
Sport (basketball)	$n = 1$	$n = 16$	3.21, n.s
Sport (Soccer)	$n = 7$	$n = 18$	
Spine injury	$n = 1$	$n = 3$	0.10, n.s
Hip injury	$n = 2$	$n = 4$	0.93, n.s
Knee injury	$n = 2$	$n = 9$	0.01, n.s
Ankle injury	$n = 6$	$n = 21$	0.49, n.s
Prior preventive training	$n = 0$	$n = 4$	1.04, n.s

Table 3 Mean ± standard deviations of baseline and changes (post-pre) biomechanical variables

	Responders ($n=8$)	Non-responders ($n=34$)	p value
Baseline			
Hip flexion ($\lambda=0.93$, n.s.)			
Peak angle (°)	56.1 ± 10.1	58.2 ± 8.8	–
Excursion (°)	33.3 ± 12.6	29.8 ± 9.7	–
Moment (Nm)	105.0 ± 29.3	103.1 ± 28.5	–
Hip adduction ($\lambda=0.82$, $p=0.049$)*			
Peak angle (°)	2.7 ± 5.9	0.9 ± 4.2	n.s
Excursion (°)	5.3 ± 4.1	2.7 ± 2.3	0.02*
Moment (Nm)	26.2 ± 22.1	17.9 ± 15.0	n.s
Knee flexion ($\lambda=0.96$, n.s.)			
Peak angle (°)	76.5 ± 3.8	80.6 ± 8.9	–
Excursion (°)	58.9 ± 6.7	61.6 ± 10.6	–
Moment (Nm)	108.1 ± 34.0	104.2 ± 24.1	–
Knee abduction ($\lambda=0.86$, n.s.)			
Peak angle (°)	12.8 ± 6.7	6.7 ± 7.0	–
Excursion (°)	11.3 ± 5.1	6.8 ± 4.6	–
Moment (Nm)	33.0 ± 14.9	22.2 ± 15.0	–
Delta scores			
Hip flexion ($\lambda=0.63$, $p<0.001$)			
Peak angle (°)	*↑ 5.1 ± 9.9*	*↓ 1.2 ± 6.0*	*0.02**
Excursion (°)	↓ 1.8 ± 7.0	↓ 0.6 ± 8.1	n.s
Moment (Nm)	*↑ 37.8 ± 31.4*	*↑ 4.2 ± 19.5*	*<0.001**
Hip adduction ($\lambda=0.95$, n.s.)			
Peak angle (°)	↓ 0.7 ± 4.4	↓ 0.9 ± 3.0	–
Excursion (°)	↑ 0.3 ± 3.0	↓ 0.4 ± 2.3	–
Moment (Nm)	↑ 5.4 ± 17.3	↑ 0.1 ± 10.2	–
Knee flexion ($\lambda=0.94$, n.s.)			
Peak angle (°)	↑ 1.8 ± 5.4	↓ 0.1 ± 2.7	–
Excursion (°)	↓ 0.1 ± 4.2	↑ 0.2 ± 6.9	–
Moment (Nm)	↑ 5.7 ± 32.4	↑ 3.9 ± 17.4	–
Knee abduction ($\lambda=0.42$, $p<0.001$)			
Peak angle (°)	↓ 6.8 ± 4.1	↑ 0.3 ± 4.6	<0.001*
Excursion (°)	↓ 5.2 ± 3.4	↓ 0.04 ± 3.9	0.001*
Moment (Nm)	↓ 17.8 ± 5.8	↑ 2.5 ± 7.2	<0.001*

*$p<0.05$

While the difference in total training sessions attended was not significant ($p=0.07$, $d=0.87$), on average, responders attended 22.5% more sessions than non-responders. Additionally, there was a trend towards a relationship between responder status and dominant sport, such that basketball players were less likely and soccer players were more likely to be responders than non-responders ($\chi^2=3.21$, $p=0.07$). No significant relationships were identified between responders and non-responders for injury history (hip, knee, ankle, and spine), past experience with preventive training, or performance variables (n.s.).

Discussion

The most important finding of the present study was that athletes who demonstrate a reduction of knee abduction moment after performing an ACL-IPP had greater hip adduction excursion at baseline, and shared greater corresponding decreases in knee abduction kinematics and increased hip flexion angles and moments. Identifying individual athletes that respond positively to an ACL-IPP via exhibiting biomechanics that place them at a lower risk of injury may help improve the efficiency of future preventive programs.

These data indicate that relative to the control group, the intervention group did not show overall reduction in knee abduction moment. Neuromuscular warm-up programs of ~20 min durations may not provide an adequate stimulus to evoke changes in knee abduction like a dedicated 60–90 min program [9]. A recent meta-analysis showed that longer weekly durations of preventive training are most successful at reducing ACL injury rates [26]. However, a recent warm-up program (FIFA 11+) was found, in a non-randomized study, to reduce knee valgus in pre-adolescent female soccer players [36]. Studies of warm-up programs report only mild improvements in knee flexion kinematics during jump landings [4, 14]. Zebis et al. [40] reported no kinematic or kinetic changes after 12 weeks of a neuromuscular warm-up program. While warm-up IPP may be more realistic for the majority of adolescent athletes, larger dedicated IPP sessions with larger stimuli during the offseason may evoke greater biomechanical changes. Additionally, heterogeneous program designs may also account for the lack of consistent improvements in lower extremity valgus after preventive training. The IPP used in this study was designed to incorporate a larger breadth of training components, leading to limited depth in any single training area. Proximal control exercises and plyometric training with technique feedback have been reported to be the most efficacious modes of reducing injury risk, with programs more focused on balance and proprioceptive training potentially less effective [24, 37]. Specific to reducing knee abduction moments, more focused technique training, specifically focused on reducing knee valgus kinematics and toe landings may drastically improve biomechanics during cutting [11]. Designing more specific ACL-IPP to achieve the desired result (i.e. greater emphasis on plyometric training to reduce knee abduction moment during landing) may ultimately improve the effectiveness of these programs.

Hip adduction excursion was the only baseline biomechanical variable that distinguished responders and non-responders. Hip adduction is an influential component of dynamic lower extremity valgus collapse. Our data would

suggest that athletes considered to have high levels of hip adduction excursion would show significant reductions in knee abduction moments after completing a standard ACL-IPP, while those with less baseline hip adduction excursion at baseline may need a more specific intervention approach. This may be because different athlete phenotypes likely exist with different risk profiles. For example, latent profile analysis have been used to identify different risk groups that respond differently to a trunk focused warmup injury prevention program [7].

The current analyses also identified a trend that soccer athletes were more likely to be responders, and basketball athletes were more likely to be non-responders. If soccer players are more likely to exhibit reduced knee abduction moments after participating in preventive training as our data indicate, this may help explain the discrepancy in success of ACL-IPPs in reducing injury between basketball and soccer players that has been reported in recent meta-analyses [15, 19]. In general, basketball players land with high knee abduction moments, and exhibit higher ground reaction forces, less hip and knee flexion excursion during double-leg landings, and increased knee abduction angles and moments during single-leg landings than soccer players, which may place them at more risk for ACL injury during jump landing activities [32]. Despite this, ACL-IPPs continue to be designed as non-specific programs that are implemented to a variety of athletes without modification. Basketball and soccer require significantly different activity demands, with a larger frequency of jumping and frontal plane activities in basketball, compared to frequent cutting and running bouts in soccer [35]. These data suggest that more specific interventions aimed at women's basketball players are needed to improve knee abduction moments and may reduce the incidence of ACL injury in this population.

In addition to reducing knee abduction moments, responders also exhibited great reductions in knee abduction angles and excursions and greater increases in hip flexion angles and moments than non-responders. Considering that the responder and non-responder groups in this study were categorized by changes in knee abduction moment, it was not surprising that responders also showed improvements in frontal plane knee kinematics. Landing in a more neutral frontal plane knee alignment would reduce the moment arm at the knee, resulting in lower forces (and moments).

Responders also exhibited greater improvements in hip flexion biomechanics. External hip flexion moments can translate to the torque produced by the hip extensors during landing. Hip extensors, specifically the gluteus maximus, work eccentrically during landing to control the femur in all three cardinal planes by slowing down hip flexion and stabilizing femoral adduction and internal rotation. Hip neuromuscular dysfunction has been implicated as a potential underlying mechanism contributing to dynamic lower extremity valgus [5, 8, 33]. Improving hip extensor muscle performance has been reported to reduce knee abduction moment during a DVJ [24], and female athletes that utilize a hip strategy (greater relative distribution of hip to knee and ankle flexion moments) exhibit less knee abduction moment during single-leg landings. Thus, increasing hip flexion moments may be one mechanism to reduce knee abduction moment during ACL-IPP.

Collectively, these findings can help clinicians better identify individuals that may not respond to ACL-IPPs and elucidate reasons for their lack of response. Prior to initiating an IPP, clinicians may be able to identify athletes with increased hip adduction excursion, using a frontal plane video with a smartphone or tablet (Fig. 1) [20]. Software programs and/or applications (e.g. Dartfish, Kinovea, Coach's Eye) can be used to estimate excursion values by taking the difference of the hip adduction angle at the instant of toe contact and at maximal inferior displacement of the center of mass. Repeated video capture during the IPP can also be used for feedback with the athlete to show changes (or lack thereof) in knee abduction and/or hip flexion kinematics. This type of screening may also be done without the use of technology, as physiotherapists can reliably identify high knee valgus angles in real-time [16]. Engaging the hip extensors during activity should be a focus in these individuals, and can be increased by utilizing components of hip focused interventions and feedback [27, 33]. Real-time biofeedback in the form of hip flexion moments or EMG can ensure athletes are activating and using their hip extensors and serve as an external visual feedback for improvement [33]. Modifying exercise prescription with exercises to activate posterior hip musculature during squat, lunging, and step-up exercise progressions will promote greater utilization of the hip extensors [5], and may ultimately help non-responders improve knee abduction moments. Additionally, data from this study indicate that clinicians need to look at biomechanical improvements at the individual level, even when working with an entire team. There is wide variability in responses to ACL-IPP and within a team there are likely to be a cohort of athletes that does not achieve the level of desired improvement.

Considering that ACL injury risk is multifactorial, these analyses are limited by classifying responders based on one biomechanical variable, when other movement strategies (e.g. stiff landings) [8, 13], biomechanical variables (e.g. timing of peak forces) [21] and anthropometric variables (e.g. joint laxity, morphology) [22, 23] may influence injury risk. However, at the present time, knee abduction moment during a drop vertical jump is the best modifiable biomechanical predictor of ACL injury risk that has been prospectively identified [8, 17, 18]. There are likely several additional anatomical and physiological independent

Fig. 1 2D representation of athlete exhibiting significant hip adduction excursion

variables which could be utilized for prediction of primary and secondary ACL injury.

Conclusion

This study indicates that hip adduction excursion biomechanics during a DVJ may help predict athletes that will respond to ACL-IPP. Further, focused intervention on improving hip extension strength and neuromuscular control to increase hip flexion external moments (hip extension internal moments) during dynamic activities should be a key strategy when attempting to reduce knee abduction moments during ACL injury prevention training.

Author contributions JBT participated in study design, data collection, training, data analysis, and manuscript preparation. AND participated in data analysis and manuscript preparation. SJS participated in study design, data analysis, and manuscript preparation. KRF participated in study design, data collection, and manuscript preparation.

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Compliance with ethical standards

Conflict of interest The authors have no conflicts of interests to declare.

Ethical approval All participants provided written informed parental consent and participant assent as approved by the Institutional Review Boards at High Point University and the University of North Carolina at Greensboro.

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